COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 3

SETS

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SET

- A ***set*** is an unordered collection of objects. Objects in a set are called ***elements***, or ***members*** of the set.
- Capital letters (e.g. A, B, C, O, N) are ordinarily used to denote sets and lowercase letters (e.g., a, b, c, p) are used to denote elements of a sets.
- Elements of a Set may NOT be the same type always!
- A set is defined only by the elements which it contains. Thus repeating an element, or changing the ordering of elements in the description of the set, does not change the set itself.
- Sets can have other sets as member.

REPRESENTATION OF SET

- Basically there are two ways of representing a set:

1) **List/Tabular form**: All elements of the set are listed, the elements being separated by commas(**,**) and are enclosed by curly braces "**{**" and "**}**"

   Example: B = { a, e, i, o, u}

   A = { 2, 4, 6, 8}

2) **Rule method, or Set builder form**: A set is defined by specifying a property that elements of the set have in common.

   Example: B = { x| x is a vowel in the English alphabets}

   A = { x| x is an even integer between 1 and 8}

TYPES OF SET

- **Empty set/ Null set**: A set with no member is called empty/null set. Empty set is denoted by ∅. Empty set is also denoted by { }

  Examples: B = { }, A = { x is a multiple of 4 | x is odd }

  - Note: ∅ ≠ **{∅}**

- **Singleton set**: A set with one element is called a singleton set. *Examples*: B = {4} , S = {a}

- **Finite set** : A set with finite number of elements in it, is called a finite set.
  Example: A = { 1, 3, 4,77}

- **Infinite set** : An infinite set is a set which contains infinite number of elements.

  Examples: N = { 0, 1, 2, 3, …….}

  B = { 1, 1/3, 1/9, 1/27, ……..}

SOME EXAMPLES OF SET

- {1, 2, 3} is the set containing elements "1" and "2" and "3."
- {1, 1, 2, 3, 3} = {1, 2, 3} since repetition is irrelevant.
- {1, 2, 3} = {3, 2, 1} since sets are unordered *(order of elements does NOT matter)*
- {0, 1, 2, 3, …} is a way we denote an infinite set (in this case, the natural numbers).
- ∅ = {} is the empty set, or the set containing no elements.
- A = { a, 2, Fred, New York } ==> Although elements are usually same type, but they may be of different types
- Example 1 : The set V of all vowels in the English alphabet cab be written as V = { a, e, i, o, u }
- Example 2: The set O of odd positive integers less than 10 can be denoted by { 1, 3, 5, 7, 9 }
- Example 4 : The set of positive integers less than 100 can be denoted by { 1, 2, 3, ……..,99 }

STANDARD NUMERICAL SETS

N = {0, 1, 2, 3, …}, natural numbers starts from zero to all positive integer

Z = {…,-2, -1, 0, 1, 2, …}, integers

$\mathbf{Z}^+$ = {1, 2, 3, …}, positive integers

R, = {-2, -1.5, 0, 1, 2.5, ….} real numbers contains all numbers of integer and fractions

∈ NOTATION

- The Greek letter "∈" (epsilon) is used to denote that an object is an *element* of a set. When crossed out "∉" denotes that the object is *not an element*."

  Example: $3 \in S$ reads: "3 is an element of the set S ".

  $3 \notin S$ reads: "3 is an not element of the set S ".

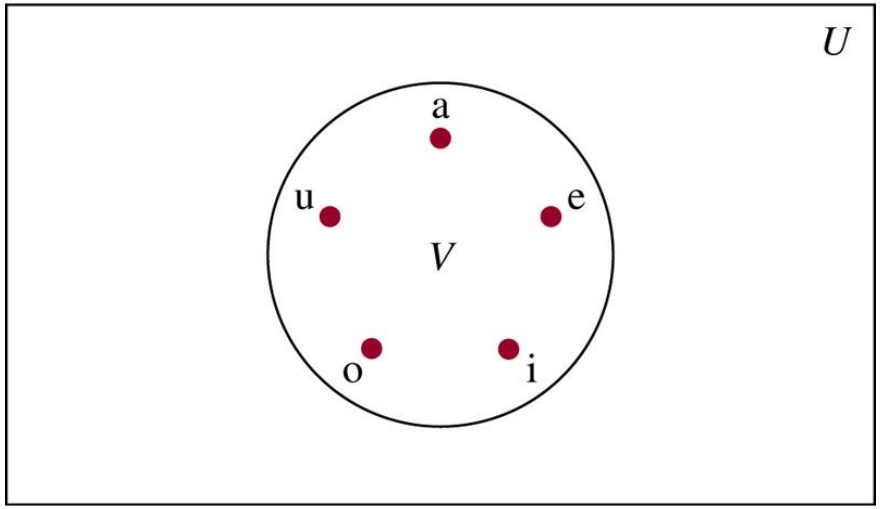
Q: Which of the following are true:

1. $3 \in \mathbf{R}$. True: 3 is a real number.
2. $-3 \in \mathbf{N}$. False: natural numbers don't contain negatives.
3. $-3 \in \mathbf{R}$. True: -3 is a real number.
4. $0 \notin \mathbf{Z}^+$. True: 0 is NOT a positive integer.
5. $\exists x\ x \in \mathbf{R} \wedge x^2 = -5$. False: square of a real number is non-negative, so can't be -5.

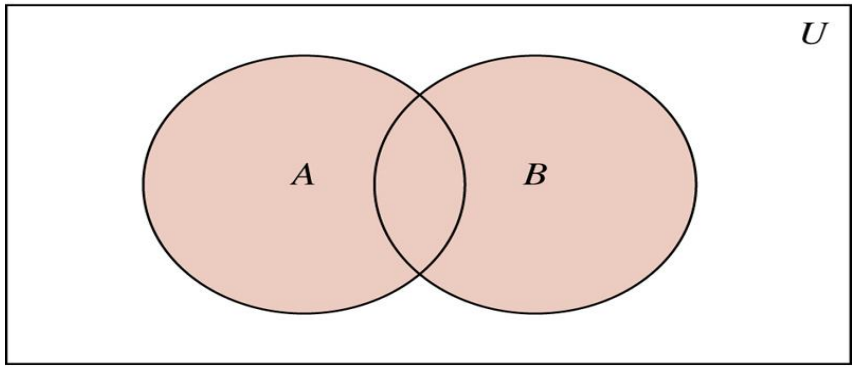
VENN DIAGRAM

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- Sets can be represented graphically using Venn diagrams.
- In Venn diagrams, the universal set U, which contains all the objects under consideration, is represented by a rectangle.
- Inside the rectangle, circles or other geometrical figures are used to represent sets. Sometimes points are used to represent the particular elements of the set.
- Venn diagrams are often used to indicate the relationships between sets.



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A *B* is shaded.

EQUAL SET

- Two sets are *equal* if and only if they have the same elements.

  $A = B$ if $\forall x(x \in A \leftrightarrow x \in B)$

- Two sets A and B are said to be equal if and only if every element of A is an element of B and consequently every element of B is an element of A; that is $A \subseteq B$ and $B \subseteq A$ and it is written as $A = B$

  Example: The sets { 1, 3, 5} & { 3, 5, 1 } are equal because they have the same elements.

SUB SET

- The set A is a subset of B if and only if every element of A is also an element of B.
- We use the notation $A \subseteq B$ to indicate that A is a subset of the set B.
- $A \subseteq B$ if and only if the quantification

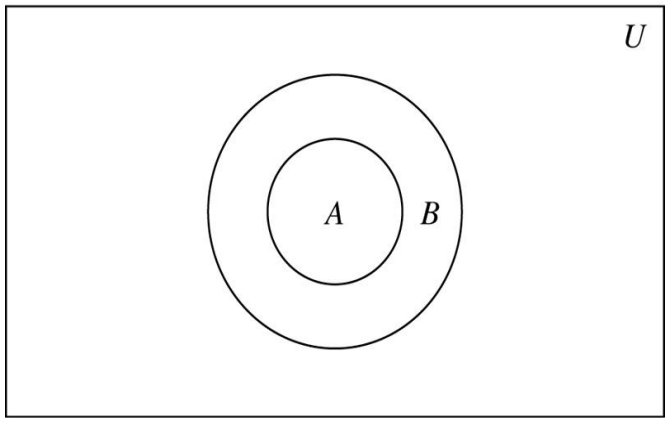
  $\forall x(x \in A \rightarrow x \in B)$ is true

- **Note**: Every nonempty set S is guaranteed to have at least two subsets, the empty set and the set S itself, that is $\varnothing \subseteq \mathbf{S}$ and $\mathbf{S} \subseteq \mathbf{S}$
- **Theorem 1**: For every set S, (1) $\varnothing \subseteq \mathbf{S}$, and (2) $\mathbf{S} \subseteq \mathbf{S}$
- ***Note***: If $A \subseteq B$ and $B \subseteq A$, then A=B
- Sets may have other sets as members

  A = {∅, {a}, {b}, { a, b} }

  B = { x | x is a subset of the set {a ,b } }

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Note: These two sets are equal, that is, A = B

Note : In the above example, **{a} ∈ A but a ∉ A**

PROPER SUBSET

- Any subset A is said to be proper subset of another set B if A is a subset of B, but there is at least one element of B which does not belong to A, i.e.,

  if $A \subseteq B$ but $A \neq B$

  $$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

- **$A \subset B$** means "A is a proper subset of B."
- **Example**: A = { 1, 5 }, B = { 1, 5, 6}, C = { 1, 6,5}

  Here, A & B both subsets of C; but A is a proper subset of C, whereas B is not a proper subset of C, since B = C

CARDINALITY OF A SET

- If there are exactly *n* distinct members in the set *S* (*n* is a non-negative integer), we say that *S* is a *finite set* and that *n* is the ***cardinality*** of *S*.
- The ***cardinality*** of a set is the number of distinct elements in the set.
- | *S* | denotes the cardinality of set *S*.
- *Examples:*

  | *S* | = *n*

  |∅| = 0

  | { 1, 5, 7, 8} | = 4

Q: Compute cardinality of each of the sets

1. | {1, -13, 4, -13, 1} | = | {1, -13, 4} | = 3
2. | {3, {1,2,3,4}, ∅} | = 3
3. | { } | = | ∅ | = 0
4. | { {}, {{}}, {{{}}} } | = | { ∅, {∅}, {{∅}} | = 3

Hint: After eliminating the redundancies just look at the number of top level commas and add 1 (except for the empty set).

CARDINALITY OF A SET

- Let A be the set of odd positive integers less than 10. Then |A| = 5 A { 1, 3, 5, 7,9}
- Let A be the set of letters in the English alphabet. Then |A| = 26 A { a, b, c, …., z}
- Because null set has no elements, it follows that, |∅| = 0
- The cardinality of the set {∅} is 1, i.e., | {∅} | = 1

- If S = {1,2,3} |S| = 3
- If S = {3,3,3,3,3} |S| = 1
- If S = {0,1,2,3,…} **|S| is infinite.**

SUPER SET

- If A is a subset of B, then B is called the super set of A and written as **$B \supseteq A$** which is read as "**B is a super set of A**".
- **$A \supseteq V$** means "**A is a superset of V**."
- Example: { 1, 4, 7, 8} is a superset of the set { 4, 7 }

- $\{1,2,3\} \subseteq \{1,2,3,4,5\}$
- $\{1,2,3\} \subset \{1,2,3,4,5\}$
  - Is $\varnothing \subseteq \{1,2,3\}$? Yes
  - Is $\varnothing \in \{1,2,3\}$? No
  - Is $\varnothing \subseteq \{\varnothing,1,2,3\}$? Yes
  - Is $\varnothing \in \{\varnothing,1,2,3\}$? Yes

- Is $\{x\} \subseteq \{x\}$? Yes
- Is $\{x\} \in \{x,\{x\}\}$? Yes
- Is $\{x\} \subseteq \{x,\{x\}\}$? Yes
- Is $\{x\} \in \{x\}$? No

POWER SET

- The *power set* of S is the set of all subsets of the set S.
- We say, "P(S) is the set of all subsets of S"
- The power set of S is denoted by ***P(S)***
- **Note**: If a set has *n* elements, then its power set has 2^n elements.

Q. What is the power set of the set {0,1,2 } ?

Solution: The power set P ({ 0, 1, 2 }) is the set of all subsets of { 0, 1, 2 } . Hence,

P ({ 0, 1, 2 }) = {∅, {0}, {1}, {2}, {0,1}, {0,2}, {1,2}, { 0,1,2}}

Q. What is the power set of empty set? What is the power set of the set {∅} ?

Solution:

P(∅) = {∅}

P({∅}) = { ∅, {∅} }

ORDERED *N*-TUPLES

- The order of elements in a collection is often important. Because sets are unordered, a different structure is needed to represent ordered collections. This is provided by **ordered *n*-tuples**.
- ***Ordered n-tuple*** $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_i as its ith element for $i = 1, 2, \ldots, n$.
- Two ordered n-tuples are equal if and only if each corresponding pair of their elements is equal. In other words, $(a_1, a_2, \ldots, a_n) = (b_1, b_2, \ldots, b_n)$

  if and only if $a_i = b_i$ for $i = 1, 2, \ldots, n$
- 2-tuples are called ***ordered pairs***. The ordered pairs (a, b) and (c, d) are equal if and only if a = c and b = d
- ***Note***: (a, b) and (b, a) are not equal unless a = b
- As opposed to sets, **repetition** and **ordering** DO MATTER with n-tuples.

  (11, 11, 11, 12, 13) ≠ (11, 12, 13)

CARTESIAN PRODUCT OF SETS

- Let A and B be sets. The *Cartesian product* of *A* and *B*, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. Cartesian product of more than two sets can be defined

  $$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

Q. What is the Cartesian product of A = { 1, 2 } and B = { a, b, c} ?

Solution: $A \times B = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c) \}$

Note : $A \times B$ and $B \times A$ are not equal, unless $A=\emptyset$ or $B=\emptyset$ or $A=B$

Q. If $A = \{1,2\}$, $B = \{3,4\}$, $C = \{5,6,7\}$ what is $A \times B \times C$?

Solution: $A \times B \times C$ = { (1,3,5), (1,3,6), (1,3,7),(1,4,5), (1,4,6), (1,4,7),
(2,3,5), (2,3,6), (2,3,7), (2,4,5), (2,4,6), (2,4,7) }

NOTATION IN SETS

- Curly brace notation "{ … }"
- Cardinality "| … |"
- Element containment ∈
- Subset containment ⊆
- Empty set { } = ∅
- Power set $P(S) = \mathbf{2^S}$
- N-tuples "(…)" and Cartesian product ×

- **Union (∪)**
- **Intersection (∩)**
- **Disjoint**
- **Difference (" – ")**
- **Complement(" ¯ ")**
- **Symmetric Difference (⊕)**

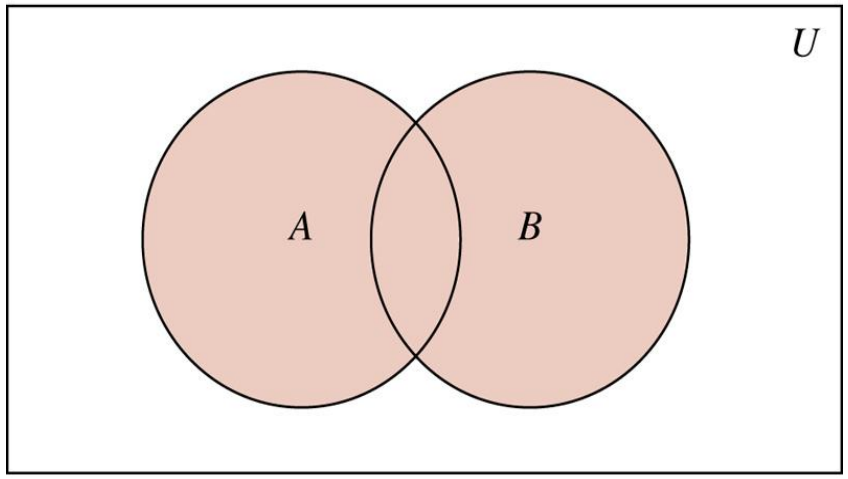
UNION OF SETS

- Let A and B be sets. The union of the sets A and B, denoted by A U B, is the set that contains those elements that are either in A or in B, or in both.
- An element x belongs to the union of the sets A and B if and only if x belongs to A or x belong to B

  $$A \cup B = \{ x \mid x \in A \vee x \in B \}$$

- A∪B is pronounced as " A union B"
- Example 1: The **union** of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 2, 3, 5}; that is, {1,3, 5} U {1, 2, 3} = {1, 2, 3, 5}
- Example 2: If A = {Cyndi, Faisal, Abdullah}, and B ={Laila, Rahim}, then, A ∪ B = {Cyndi, Faisal, Abdullah, Laila, Rahim}

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A B is shaded.

FIGURE: Venn Diagram Representing The Union Of *A* And *B*.

INTERSECTION OF SETS

- Let ***A*** and ***B*** be sets. The *intersection* of the sets *A* and *B*, denoted by $A \cap B$, is the set containing those elements in both *A* and *B*.
- An elements x belongs to the *intersection* of the sets *A* and *B* if and only if x belongs to A **and** x belongs to B

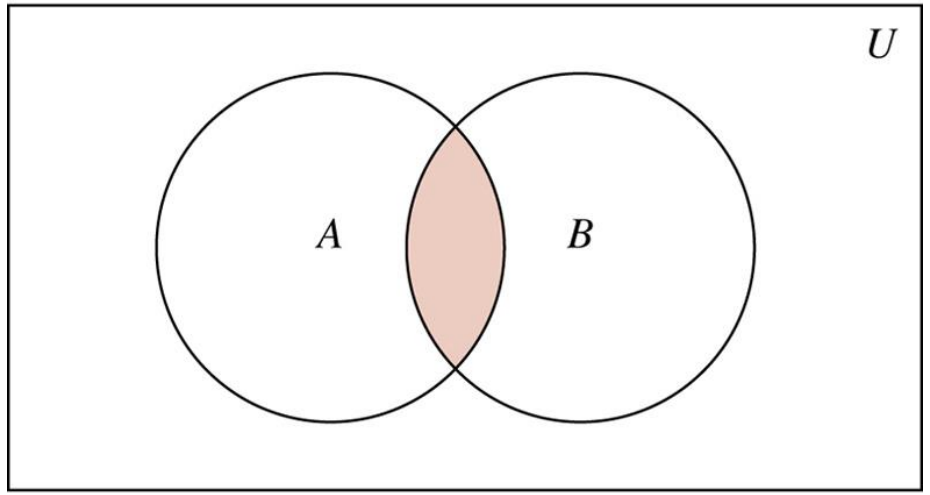
$$A \cap B = \{ x \mid x \in A \wedge x \in B \}$$

Example 1: The *intersection* of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 3}; that is, $\{1,3,5\} \cap \{1,2,3\} = \{1,3\}$

Example 2: If A = {x : x is a Bangladeshi president}, and B = {x : x is deceased}, then

$A \cap B$ = {x : x is a deceased Bangladeshi president}

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A *B* is shaded.

FIGURE: Venn Diagram Representing The Intersection Of *A* And *B*.

UNION AND INTERSECTION OF THREE SETS

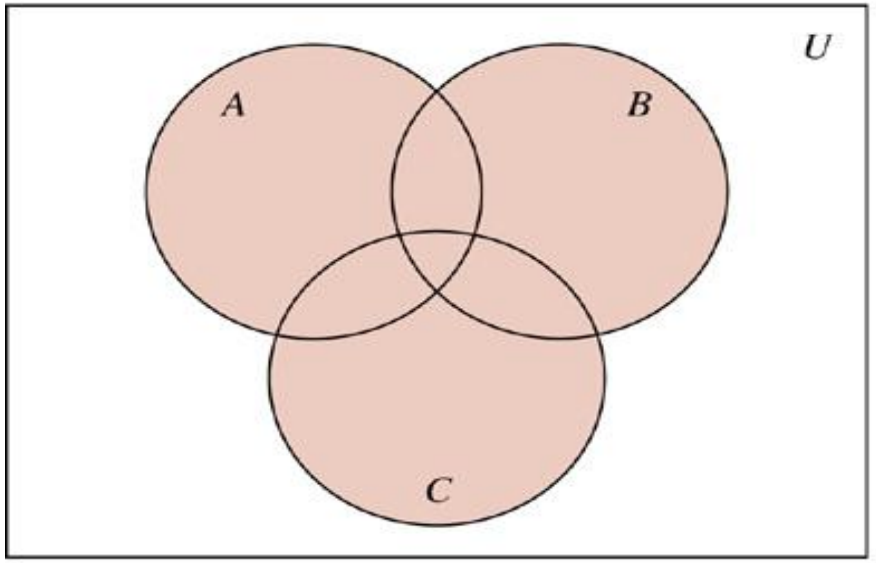
Ex: Let A = { 0, 2, 4, 6, 8}, B = {0, 1, 2, 3, 4}, and C = { 0, 3, 6, 9}. What are A U B U C and A ∩ B ∩ C ? Show the above sets in a Venn Diagram. Assume universal set U.

Solution:

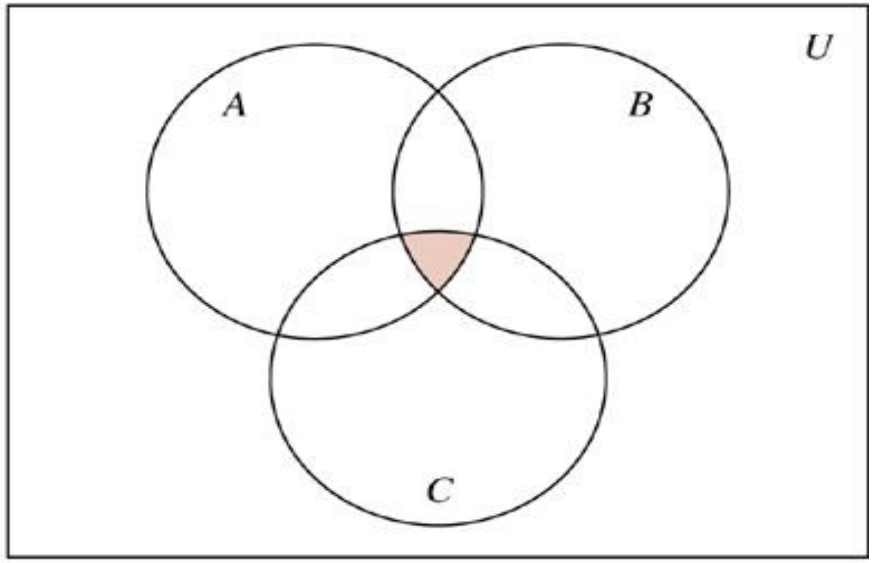
A U B U C = { 0, 1, 2, 3, 4, 6, 8, 9}

A ∩ B ∩ C = { 0 }

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(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

DISJOINT SETS

- Two sets are ***disjoint*** if their intersection is the empty set.
- Let A and B be sets. **If $A \cap B = \varnothing$** , that is if A and B do not have any element(s) in common, then A and B are said to be disjoint or non-intersecting.
- *Disjoint ==> Non-intersecting*
- Example: Let *A*= {1, 3, 5, 7, 9} and *B*={2, 4, 6, 8, 10}. Because ***$A \cap B = \varnothing$***, A and B are ***disjoint***.

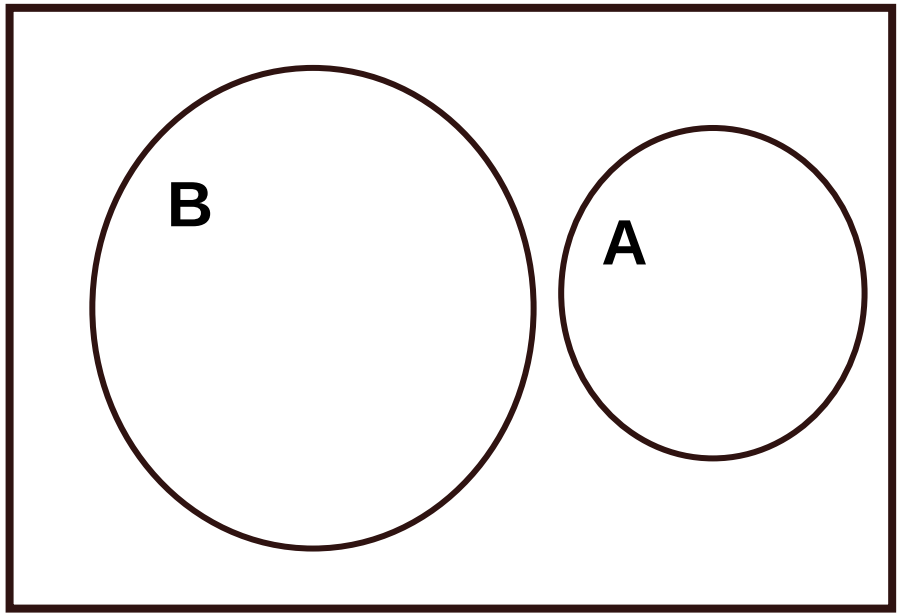


FIGURE: Venn Diagram Showing Disjoint Of The Two Sets A And B

Principle of inclusion-exclusion

Q: What is the ***cardinality*** of a union of two sets A and B?

Answer: **$|A \cup B| = |A| + |B| - |A \cap B|$**

DIFFERENCE OF TWO SETS

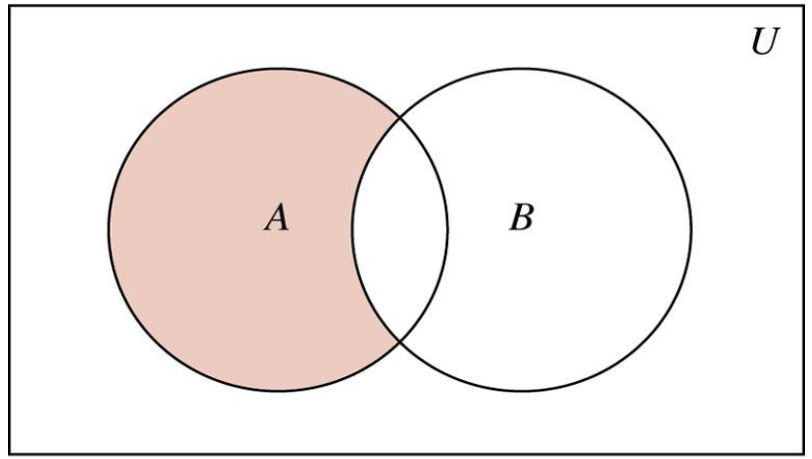
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- The ***difference*** of the sets *A* and *B*, denoted by *A* – *B*, is the set containing those elements that are in *A* but not in *B*.
- An element x belongs to the difference of A and B if and only if $x \in A$ **and** $x \notin B$. This tell us that

$$A - B = \{ x \mid x \in A \wedge x \notin B \}$$

- The difference of A and B is also called the *complement of B with respect to A*.

$$A - B = A \cap \overline{B}$$



$A - B$ is shaded.

FIGURE: Venn Diagram for the Difference of *A* and *B*.

- Ex: The difference of {1, 3, 5} and {1, 2, 3} is the set {5}; that is, {1, 3, 5} – {1, 2, 3} = {5}.
- Note: This is the different from the difference of {1, 2, 3} and {1, 3, 5} which is the set {2}.
- Ex: Let A = { a, b, c} and B = { b, c, d, e }.

$A - B = \{ a \}$
$B - A = \{ d, e \}$

COMPLEMENT OF A SETS

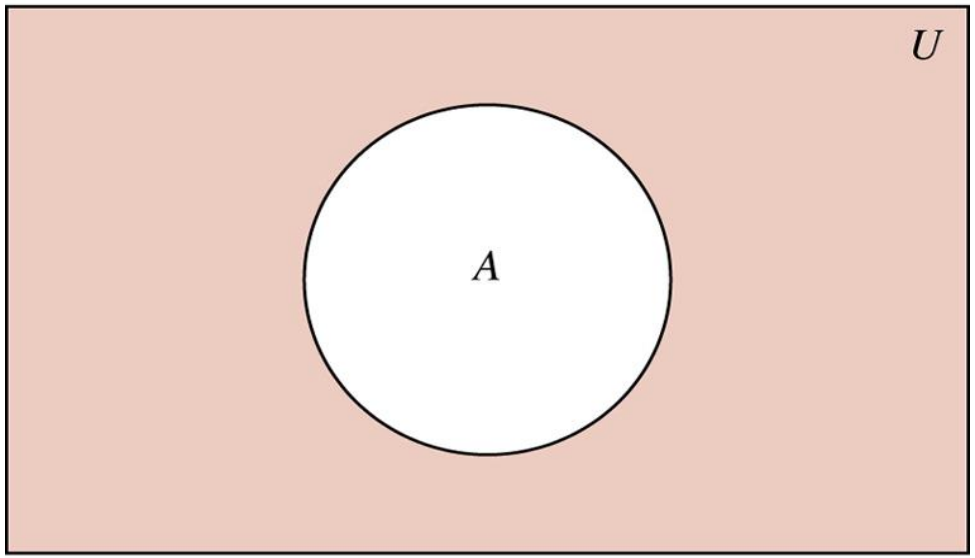
- Let *U* be the universal set. The ***complement*** of the set *A*, denoted by $\bar{A}$, is the complement of *A* with respect to *U*.
- An element belongs to $\bar{A}$ if and only if $x \notin A$

  $\bar{A} = \{ x \mid x \notin A \}$
- In other words, the complement of the set A is ***U – A,***

  $\bar{A} = U - A$

$$\bar{A} = \{ x \mid x \notin A \}$$

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$\bar{A}$ is shaded.

FIGURE : Venn Diagram For The Complement Of The Set *A*

SYMMETRIC DIFFERENCE OF TWO SETS

- The symmetric difference of two sets *A* and *B*, denoted by $A \oplus B$, is the set of elements that belongs to *A* or to *B*, but not to both *A* and *B*.

  A ⊕ B = The set containing those elements in exactly one of A and B.

  $$\mathbf{A \oplus B = \{ x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) \}}$$

  $$= \boldsymbol{(A - B) \cup (B - A)}$$

- Example: If A = { -3, 0, 1, 2} and B = { 1, 2, 3, 4 }, then A ⊕ B = { -3, 0, 3, 4}

  Alternate Solution: A – B = { -3, 0 } and B – A = { 3, 4 }. Therefore, (A - B) U (B - A) = { -3, 0, 3, 4 }

  $$\mathbf{A \oplus B =} \{ x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

  $$= \{ x \mid (x \in A - B) \vee (x \in B - A)\}$$

  $$= \{ x \mid x \in ((A - B) \cup (B - A))\}$$

  $$= (A - B) \cup (B - A)$$

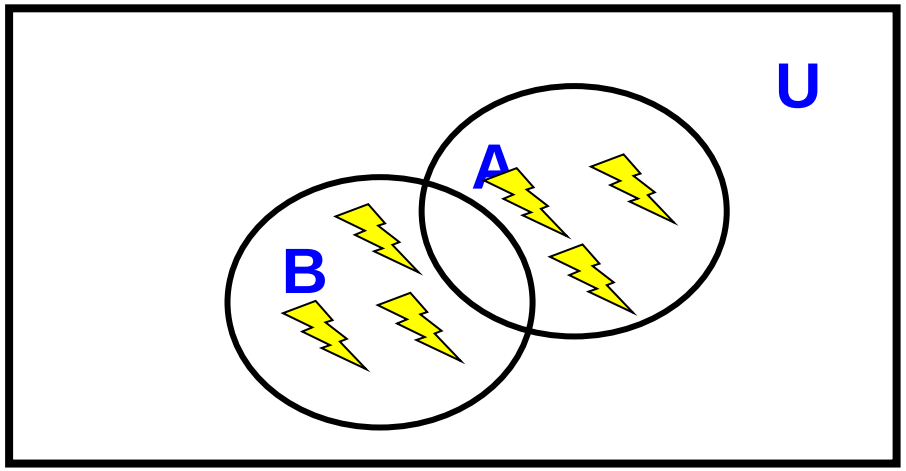


Figure: Venn Diagram For The *Symmetric Difference Of Two Sets A And B*, A ⊕ B

SET IDENTITIES

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TABLE 1 Set Identities.

Identity	*Name*
$A \cup \varnothing = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \varnothing = \varnothing$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \varnothing$	Complement laws

- ***Disjunction*** **"∨" becomes** ***union*** **"∪"**
- ***Conjunction*** **"∧" becomes** ***intersection*** **"∩"**
- ***Negation*** **"¬" becomes** ***complementation*** **"–"**
- ***False*** **"F" becomes** ***the empty set*** **∅**
- ***True*** **"T" becomes** ***the universal set*** **U**

SET IDENTITY: USING MEMBERSHIP TABLE

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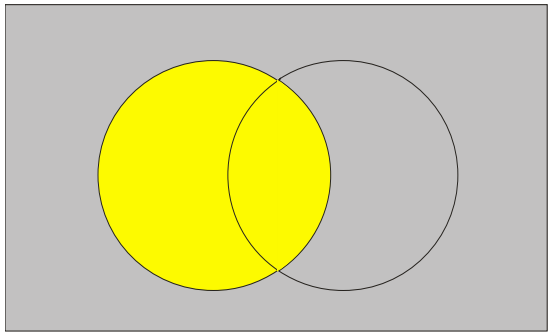
TABLE 2 A Membership Table for the Distributive Property.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

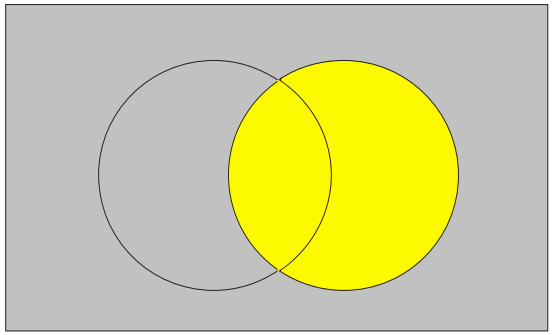
SET IDENTITIES VIA VENN DIAGRAM

- De Morgan's first law can be visualized as follows. $\overline{A \cup B} = \overline{A} \cap \overline{B}$

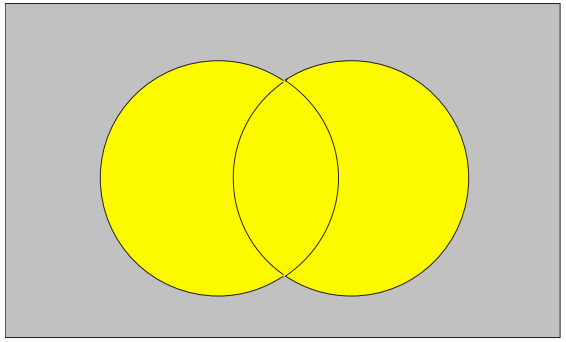
A:



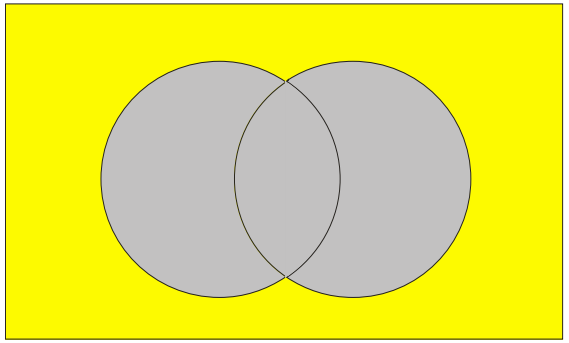
B:



$A \cup B$:



$\overline{A \cup B}$:



SET IDENTITIES

Prove that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ using a membership table/ set builder notation and logically equivalences

0 : x is not in the specified set

1 : otherwise

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cap \overline{B}$	A U B	$\overline{A \cup B}$
1	1	0	0	0	1	0
1	0	0	1	0	1	0
0	1	1	0	0	1	0
0	0	1	1	1	0	1

$(A \cup B) = \{x \mid \neg (x \in A \vee x \in B)\}$

$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$

$= \{x \mid (x \in \overline{A}) \wedge (x \in \overline{B})\}$

$= \overline{A} \cap \overline{B}$

REFERENCES

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- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley